Customer Data Analysis Project Report

1. Project Overview

This project analyses customer shopping transactions from 10 shopping malls in Istanbul between 2021 and 2023. The analysis explores customer shopping patterns across gender, age, product categories and payment methods. SQL is used to examine the dataset, assess data quality and calculate transaction, quantity and revenue measures. The findings are intended to support business understanding and data-driven decisions.

2. Dataset Summary

Dataset Measure	Details
Total Rows	99,457
Total Columns	10
Key Features	Invoice number, Customer ID, Gender, Age, Category, Quantity, Price, Payment Method, Invoice Date and Shopping Mall
Shopping Malls	10
Data Period	2021–2023
Missing Values	0 NULL values reported across all 10 columns
Duplicate Invoice Numbers	No duplicate invoice numbers returned by the duplicate-check query

3. Exploratory Data Analysis Using SQL

The exploratory analysis began by checking the available databases and tables, inspecting the customer table structure, and confirming the dataset size. Data quality checks were then performed to identify NULL values and repeated invoice numbers. The initial checks and their outputs are documented in the following sections.

3.1 Database and Table Exploration

The project database and the customer table were inspected using database and table discovery queries.

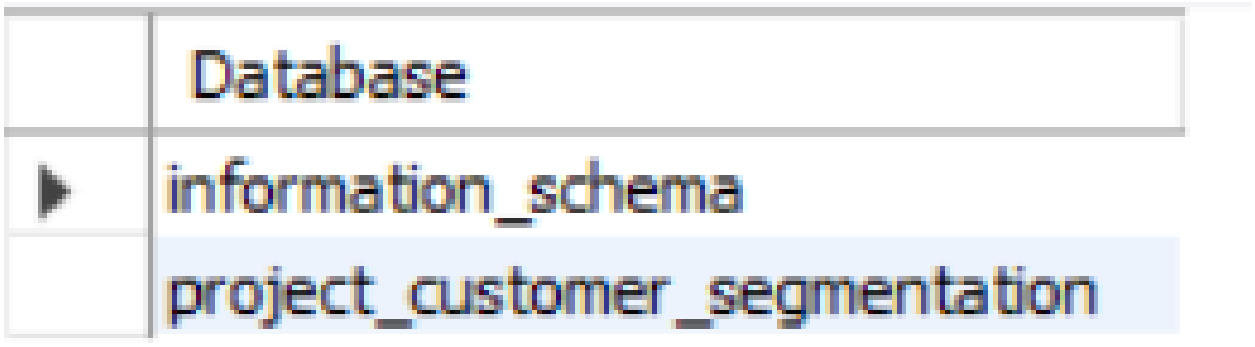


Figure 1. Available databases displayed using SHOW DATABASES.

	Tables_in_project_customer_segmentation
▶	customer

Figure 2. Tables available in the selected database.

3.2 Customer Table Structure

The DESCRIBE query was used to inspect the 10 fields and their MySQL data types.

	Field	Type	Null	Key	Default	Extra
▶	invoice_no	varchar(7)	NO	PRI	NULL	
	customer_id	varchar(7)	NO		NULL	
	gender	varchar(6)	NO		NULL	
	age	int(11)	NO		NULL	
	category	varchar(15)	NO		NULL	
	quantity	int(11)	NO		NULL	
	price	decimal(7,2)	NO		NULL	
	payment_method	varchar(11)	NO		NULL	
	invoice_date	varchar(16)	NO		NULL	
	shopping_mall	varchar(17)	NO		NULL	

Figure 3. Customer table structure returned by DESCRIBE customer.

3.3 Dataset Size

The record-count query returned 99,457. This is the total number of records and therefore the total number of rows in the customer table. The column-count query returned 10.



Figure 4. Total record count: 99,457 rows.

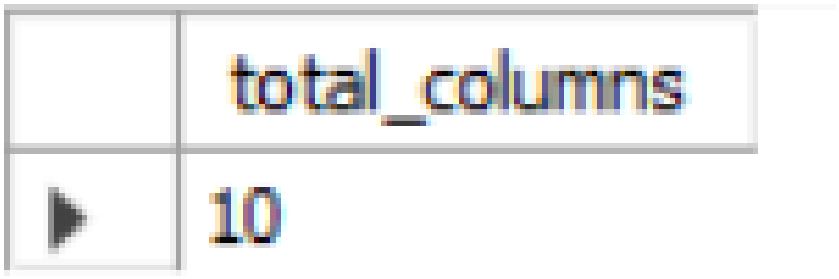


Figure 5. Total column count: 10.

3.4 Missing Value Identification

A NULL-value check was performed for each column. The supplied output shows a NULL count of 0 for all 10 columns. Based on this check, no missing values were identified in the dataset.

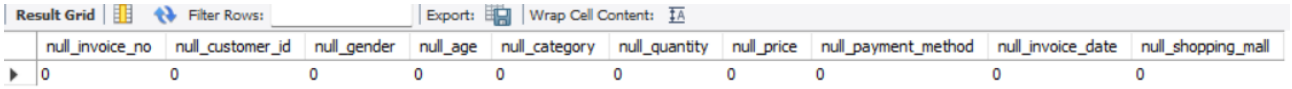


Figure 6. NULL-value check showing zero NULL values in all 10 columns.

3.5 Duplicate Invoice Identification

A duplicate check was performed on invoice_no by grouping records and filtering for invoice numbers that occurred more than once. The query returned no data rows, indicating that no duplicate invoice numbers were identified.



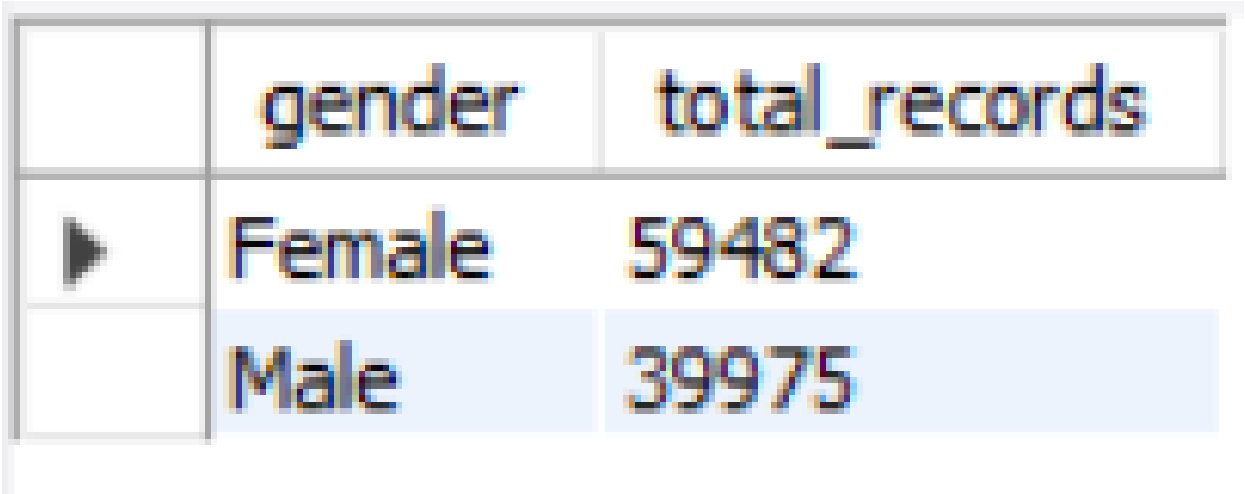
Figure 7. Duplicate invoice query returned no matching duplicate rows.

4. SQL-Based Customer Data Analysis

The following sections present the SQL analysis outputs supplied so far. Where an output screenshot was not available because the result contained many rows, the documented key results are summarized in text instead of reproducing a partial screenshot.

4.1 Shopping Distribution by Gender

Female customers accounted for 59,482 transactions (59.81%), while male customers accounted for 39,975 transactions (40.19%). Female customers represented the larger share of shopping transactions in the dataset.

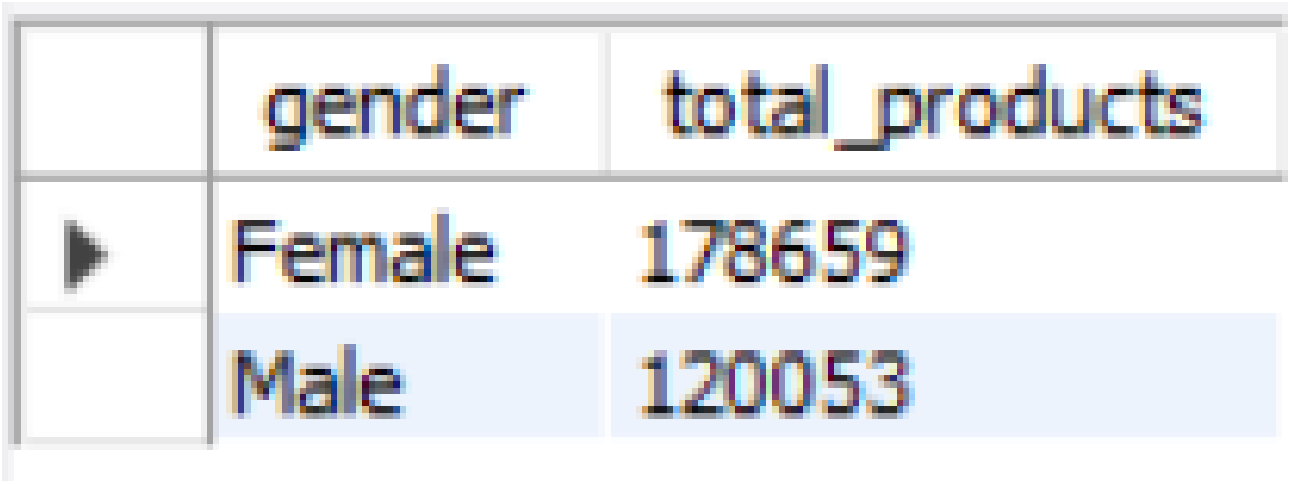


gender	total_records
Female	59482
Male	39975

Figure 8. SQL output: shopping distribution by gender.

4.2 Product Quantity by Gender

Female customers purchased 178,659 products (59.82%), compared with 120,053 products (40.18%) purchased by male customers. The total quantity sold was 298,712 products.

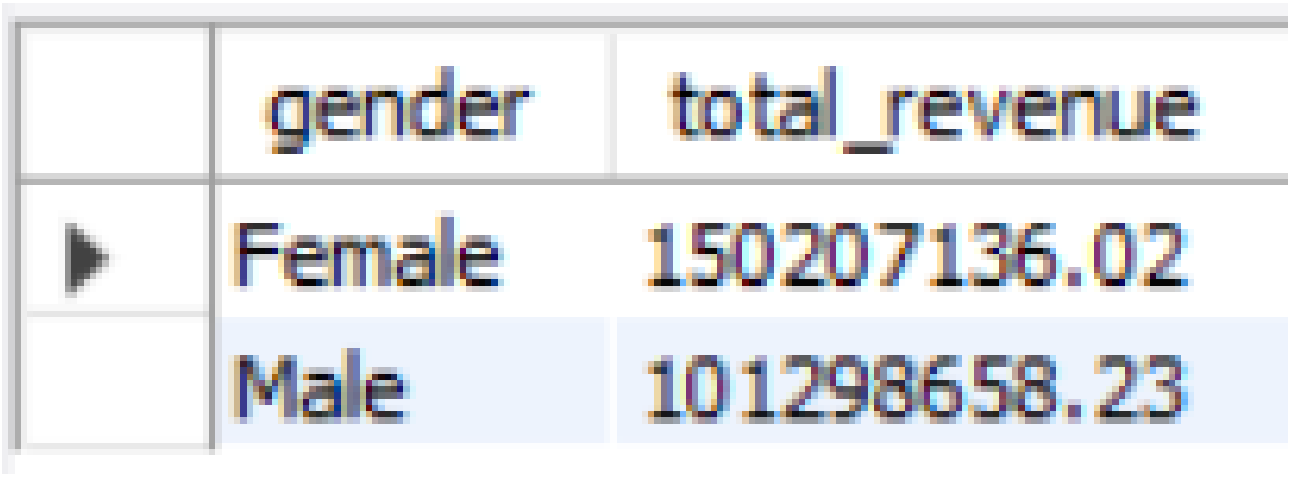


gender	total_products
Female	178659
Male	120053

Figure 9. SQL output: total product quantity by gender.

4.3 Revenue by Gender

Female customers generated 150,207,136.02 in revenue (59.72%), while male customers generated 101,298,658.23 (40.28%). Total revenue was 251,505,794.25.



gender	total_revenue
Female	150207136.02
Male	101298658.23

Figure 10. SQL output: revenue by gender.

4.4 Purchase Category Distribution

Clothing recorded the highest transaction count with 34,487 records (34.68%). Cosmetics followed with 15,097 (15.18%), and Food & Beverage recorded 14,776 (14.86%). Toys, Shoes, Souvenir, Technology and Books accounted for the remaining transactions.

category	total_records
Books	4981
Clothing	34487
Cosmetics	15097
Food & Beverage	14776
Shoes	10034
Souvenir	4999
Technology	4996
Toys	10087

Figure 11. SQL output: overall purchase category distribution.

4.4A Purchase Category Distribution by Gender

Female customers recorded more transactions than male customers across all eight product categories. Clothing was the leading category for both genders, with 20,652 female transactions and 13,835 male transactions.

category	gender	total_records
Books	Female	2906
Books	Male	2075
Clothing	Female	20652
Clothing	Male	13835
Cosmetics	Female	9070
Cosmetics	Male	6027
Food & Beverage	Female	8804
Food & Beverage	Male	5972
Shoes	Female	5967
Shoes	Male	4067
Souvenir	Female	3017
Souvenir	Male	1982
Technology	Female	2981
Technology	Male	2015
Toys	Female	6085
Toys	Male	4002

Figure 12. SQL output: category distribution by gender.

4.4B Purchase Category Distribution by Age

The documented analysis identifies Clothing as the leading purchase category across customer ages. Customers aged 37 recorded the highest transaction count (2,057), while customers aged 54 recorded the lowest (1,830). Customers aged 18–29 contributed 23,043 transactions, approximately 23.17% of all transactions. The full age-by-category output contains many rows, so the documented results are summarized here.

4.4C Purchase Category Distribution by Payment Method

Cash was the most frequently used payment method overall, accounting for 44.69% of transactions, followed by Credit Card at 35.12% and Debit Card at 20.19%. Clothing recorded the highest transaction count across all three payment methods.

category	payment_method	total_records
Clothing	Cash	15456
Clothing	Credit Card	12025
Clothing	Debit Card	7006
Cosmetics	Cash	6674
Food & Beverage	Cash	6587
Cosmetics	Credit Card	5336
Food & Beverage	Credit Card	5250
Toys	Cash	4542
Shoes	Cash	4474
Shoes	Credit Card	3553
Toys	Credit Card	3548
Cosmetics	Debit Card	3087
Food & Beverage	Debit Card	2939
Books	Cash	2268
Technology	Cash	2235
Souvenir	Cash	2211
Shoes	Debit Card	2007
Toys	Debit Card	1997
Souvenir	Credit Card	1770
Technology	Credit Card	1753
Books	Credit Card	1696
Souvenir	Debit Card	1018
Books	Debit Card	1017
Technology	Debit Card	1008

Figure 13. SQL output: category distribution by payment method.

4.4D Purchase Category Distribution by Shopping Mall

Mall of Istanbul recorded 19,943 transactions (20.05%), followed by Kanyon with 19,823 (19.93%) and Metrocity with 15,011 (15.09%). Emaar Square Mall recorded the lowest transaction count, with 4,811 (4.84%). Clothing was the leading category across all 10 malls. The full category-by-mall output is summarized in text because it contains many rows.

4.5 Shopping Distribution by Age

Shopping transactions were recorded across customers aged 18 to 69. Customers aged 37 recorded the highest number of transactions (2,057), while customers aged 54 recorded the lowest (1,830). Customers aged 18–29 contributed 23,043 transactions, approximately 23.17% of the total. The full age-wise output contains many rows, so the documented results are summarized here.

4.6 Product Quantity by Age

Customers aged 37 purchased the highest quantity, with 6,217 products. Customers aged 54 purchased the lowest quantity, with 5,471 products. Customers aged 22 ranked second with 6,148 products. Across all ages, the total quantity sold was 298,712 products. The full age-wise output is summarized in text because of its row count.

4.7 Revenue by Age

Customers aged 37 generated the highest revenue, approximately 5.58 million, followed by customers aged 64 at approximately 5.27 million and customers aged 51 at approximately 5.24 million. Customers aged 18 generated the lowest revenue, approximately 4.40 million. The complete age-wise output is summarized in text rather than shown as a partial screenshot.

4.9A Payment Method Distribution by Gender

Female customers recorded higher transactions than male customers across all three payment methods. Cash was the most frequently used payment method for both genders. Female customers recorded 26,509 cash, 21,011 credit card and 11,962 debit card transactions; male customers recorded 17,938 cash, 13,920 credit card and 8,117 debit card transactions.

	gender	payment_method	total_records
▶	Female	Cash	26509
	Female	Credit Card	21011
	Female	Debit Card	11962
	Male	Cash	17938
	Male	Credit Card	13920
	Male	Debit Card	8117

Figure 14. SQL output: payment method distribution by gender.

4.9B Payment Method Distribution by Age

Cash was the most preferred payment method across all 52 customer ages. Customers aged 37 recorded the highest total transactions (2,057), while customers aged 54 recorded the lowest (1,830). Overall, Cash accounted for 44.69% of transactions, Credit Card 35.12% and Debit Card 20.19%. The full age-by-payment output contains many rows, so the documented results are summarized in text.

4.9C Payment Method Distribution by Category

Cash was the most commonly used payment method across all eight product categories. Clothing recorded the highest transactions across payment methods, with 15,456 cash, 12,025 credit card and 7,006 debit card transactions. Credit Card ranked second overall, while Debit Card was the least-used method.

category	payment_method	total_records
Books	Cash	2268
Books	Credit Card	1696
Books	Debit Card	1017
Clothing	Cash	15456
Clothing	Credit Card	12025
Clothing	Debit Card	7006
Cosmetics	Cash	6674
Cosmetics	Credit Card	5336
Cosmetics	Debit Card	3087
Food & Beverage	Cash	6587
Food & Beverage	Credit Card	5250
Food & Beverage	Debit Card	2939
Shoes	Cash	4474
Shoes	Credit Card	3553
Shoes	Debit Card	2007
Souvenir	Cash	2211
Souvenir	Credit Card	1770
Souvenir	Debit Card	1018
Technology	Cash	2235
Technology	Credit Card	1753
Technology	Debit Card	1008
Toys	Cash	4542
Toys	Credit Card	3548
Toys	Debit Card	1997

Figure 15. SQL output: payment method distribution by category.

4.9D Payment Method Distribution by Shopping Mall

Cash was the most preferred payment method across all shopping malls, followed by Credit Card and Debit Card. Mall of Istanbul recorded 19,943 transactions, followed by Kanyon with 19,823 and Metrocity with 15,011. Emaar Square Mall recorded the lowest total with 4,811 transactions. The full mall-by-payment output is summarized in text because it contains multiple rows.

4.10 Overall Payment Method Distribution

Cash was the most preferred payment method, with 44,447 transactions (44.69%). Credit Card ranked second with 34,931 transactions (35.12%), while Debit Card recorded 20,079 transactions (20.19%). The dataset contained 99,457 transactions in total.

payment_method	total_records
Cash	44447
Credit Card	34931
Debit Card	20079

Figure 16. SQL output: overall payment method distribution.

5. Power BI Dashboard

The Power BI dashboard brings together key customer and sales measures, including total transactions, total products sold, total revenue and unique customers. It also presents interactive filters for gender, product category and shopping mall, alongside visuals showing category distribution, shopping activity by gender and age, revenue by gender, and payment method patterns.

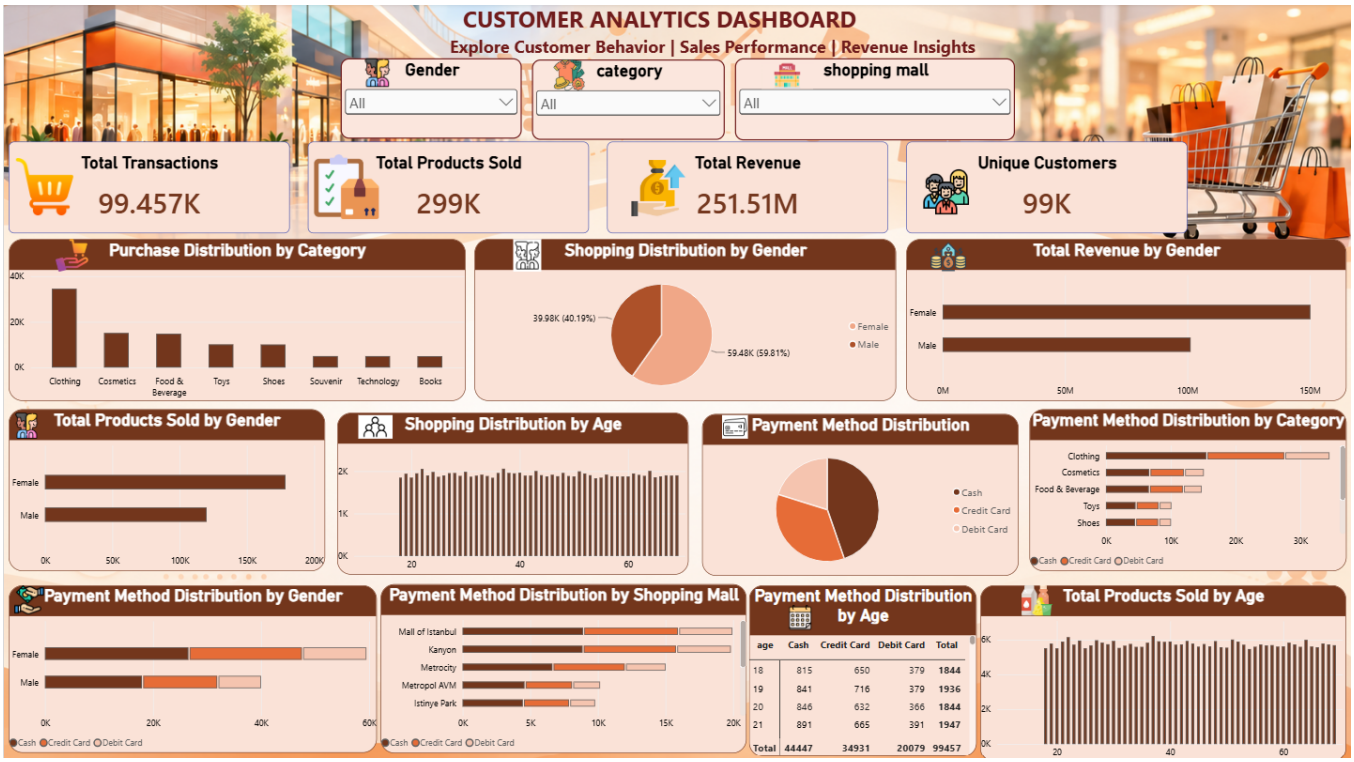


Figure 17. Customer Analytics Dashboard developed in Microsoft Power BI.